D. Y. Patil Agricultural and Technical University

Talsande

PROJECT REPORT ON

Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization

Department

Master Of Computer Applications

(MCA-Sandwich Sem-II)

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INTRODUCTION

1.1 Project Overview

In today's highly competitive business landscape, data-driven analysis is essential for optimizing sales and marketing strategies. A critical component of this optimization is strategic product placement, which influences customer purchasing behavior by positioning products to maximize visibility and consumer attraction.

This project focuses on analyzing historical sales data to evaluate the direct impact of product placement on overall sales performance. By leveraging advanced data visualization tools like Tableau, raw datasets are transformed into interactive dashboards and actionable insights. These visualizations enable businesses to identify emerging trends, seasonal patterns, and evolving customer preferences with precision.

Through comprehensive analysis categorized by product type, geographical region, and time intervals, this project provides a clear understanding of product performance. The ultimate goal is to provide businesses with a data-backed framework to refine their placement strategies and drive sustainable revenue growth.

1.2 Purpose

The purpose of this project is to analyse sales data and determine how product placement strategies affect sales performance. The project uses Tableau to create interactive dashboards that help visualize relationships between products, locations, and customer buying behaviour.

The main goal is to provide insights that help businesses make better decisions regarding product placement, marketing strategies, and sales planning. By understanding which products generate higher sales and where they perform best, organizations can optimize their product positioning and improve overall business performance.

IDEATION PHASE

2.1 Problem Statement

Many businesses collect large amounts of sales data but struggle to extract useful insights from it. Without proper analysis, companies cannot clearly understand which products are performing well or how product placement influences sales.

As a result, businesses may make decisions based on assumptions rather than data-driven insights. This may lead to inefficient product positioning, reduced sales performance, and missed business opportunities.

This project aims to solve this problem by analysing sales data and presenting insights using Tableau dashboards. These visualizations help identify sales patterns and evaluate the effectiveness of product placement strategies.

2.2 Empathy Map Canvas

The empathy map helps understand the needs and expectations of stakeholders such as business managers, marketing teams, and analysts.

Think:
Managers want to improve product visibility and increase sales.

Feel:
They feel uncertain about which product placement strategies work best.

Say:
They often say that better insights from data can help them make better business decisions.

Do:
They analyse sales reports, monitor product performance, and adjust marketing strategies.

Pain:
Large datasets are difficult to analyse manually, and valuable insights may be missed.

Gain:
Interactive dashboards and visual analytics help them quickly understand sales patterns and optimize product placement strategies.

2.3 Brainstorming

During brainstorming, several ideas were generated to improve the understanding of product placement and sales performance.

Some of the main ideas include analysing sales trends by product category, studying regional sales performance, identifying best-selling products, and visualizing monthly or yearly sales growth. Another idea is to analyse profit margins and compare them with sales performance to understand which products are most profitable.

Using Tableau dashboards makes it easier to present these insights visually so that decision-makers can quickly interpret the data and take appropriate action.

REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey map represents the steps involved in analysing sales data and generating insights.

1. The process begins with collecting sales data from available datasets.
2. The collected data is cleaned and prepared for analysis by removing errors and missing values.
3. The processed dataset is then imported into Tableau.
4. Various visualizations such as bar charts, line charts, and dashboards are created.
5. Users interact with the dashboard to explore sales trends and patterns.
6. Based on these insights, businesses can improve product placement strategies.

3.2 Solution Requirement

Functional Requirements

- Import and manage sales datasets.
- Create interactive dashboards using Tableau.
- Visualize sales performance across product categories.
- Analyse regional and time-based sales trends.
- Provide filters and interactive features for deeper analysis.

Non-Functional Requirements

- The system should be easy to use and understand.
- Dashboards should display data quickly and accurately.
- Visualizations should be clear and informative.
- The system should support large datasets efficiently.

Data Flow Diagram

Level 0 DFD

User → Dataset → Tableau Visualization → Sales Insights → Decision Making

Level 1 DFD

Dataset → Data Cleaning → Data Processing → Tableau Dashboard → Business Analysis

Technology Stack

Technology	Description
Tableau 2026.1	Used for data visualization and dashboard creation
Excel	Used as the dataset source
Python (Flask)	Used for creating web app
GitHub	Used for storing project files and version control
Web Browser	Used to access dashboards and project repository

PROJECT DESIGN

4.1 Problem Solution Fit

The project addresses the problem of understanding product placement effectiveness by converting raw sales data into visual insights. Tableau dashboards allow users to easily interpret sales trends and identify which products generate the most revenue. This helps businesses make informed decisions about product placement strategies.

4.2 Proposed Solution

The proposed solution is to create an interactive data visualization dashboard using Tableau. The dashboard will display various sales metrics such as product category performance, regional sales distribution, profit analysis, and sales trends over time.

These visualizations help users analyse patterns and identify opportunities to improve product placement strategies.

4.3 Solution Architecture

Data Source (Sales Dataset)

↓

Data Cleaning and Preparation

↓

Tableau Visualization and Dashboard Creation

↓

Sales Analysis and Insights

↓

Business Decision Making

PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was divided into several phases to ensure proper implementation.

1. Data collection from online datasets.
2. Data cleaning and preprocessing.
3. Importing dataset into Tableau.
4. Creating charts and dashboards.
5. Testing and validating visualizations.
6. Publishing the project on GitHub and Tableau Public.

FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

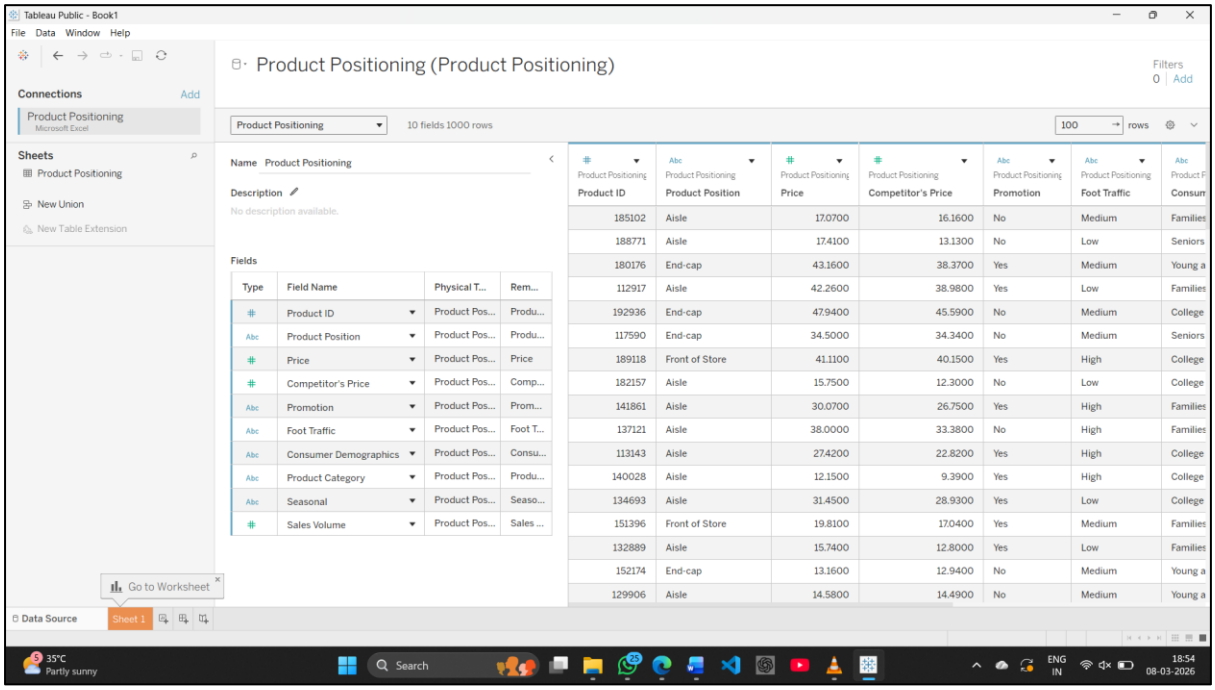
Test Case	Expected Result
Load dataset	Dataset loads successfully
Create visualization	Charts display correct values
Apply filters	Dashboard updates dynamically
Analyse trends	Sales patterns are clearly visible

RESULTS

7.1 Output Screenshots

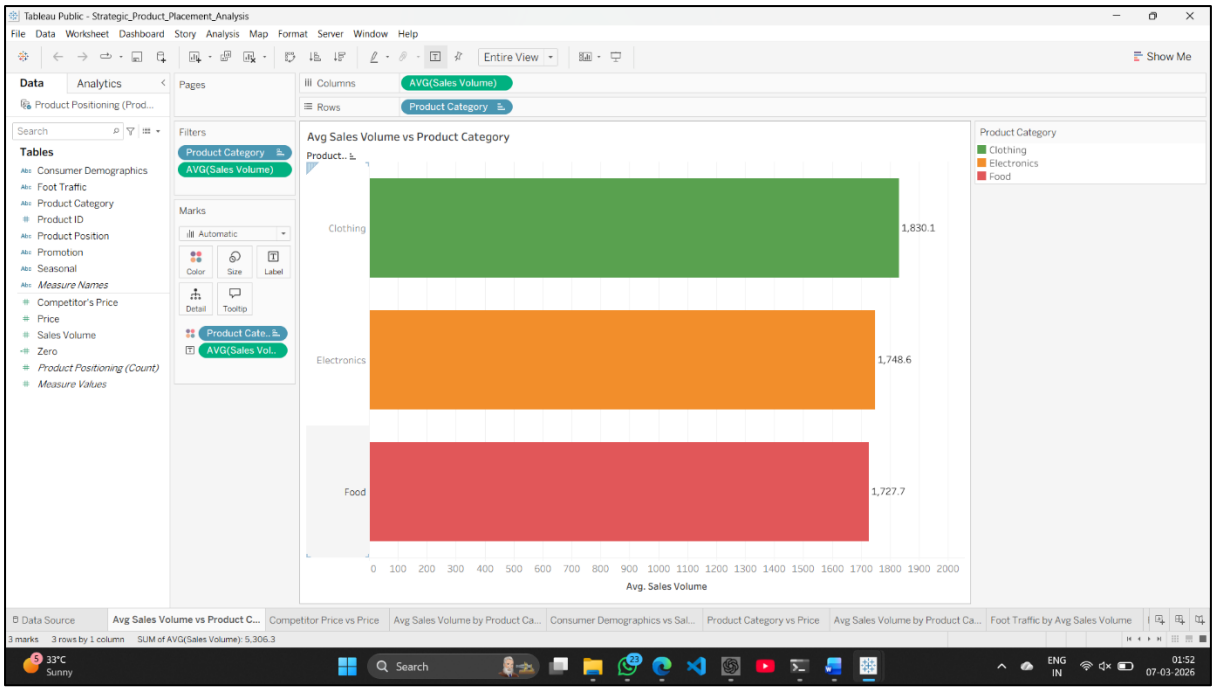
The project generates interactive dashboards that display various insights such as sales trends, product category performance, and regional sales distribution. Screenshots of the Tableau dashboard, charts, and analysis results are included in this section to demonstrate the output of the project.

- **Data Collection and Extraction:**

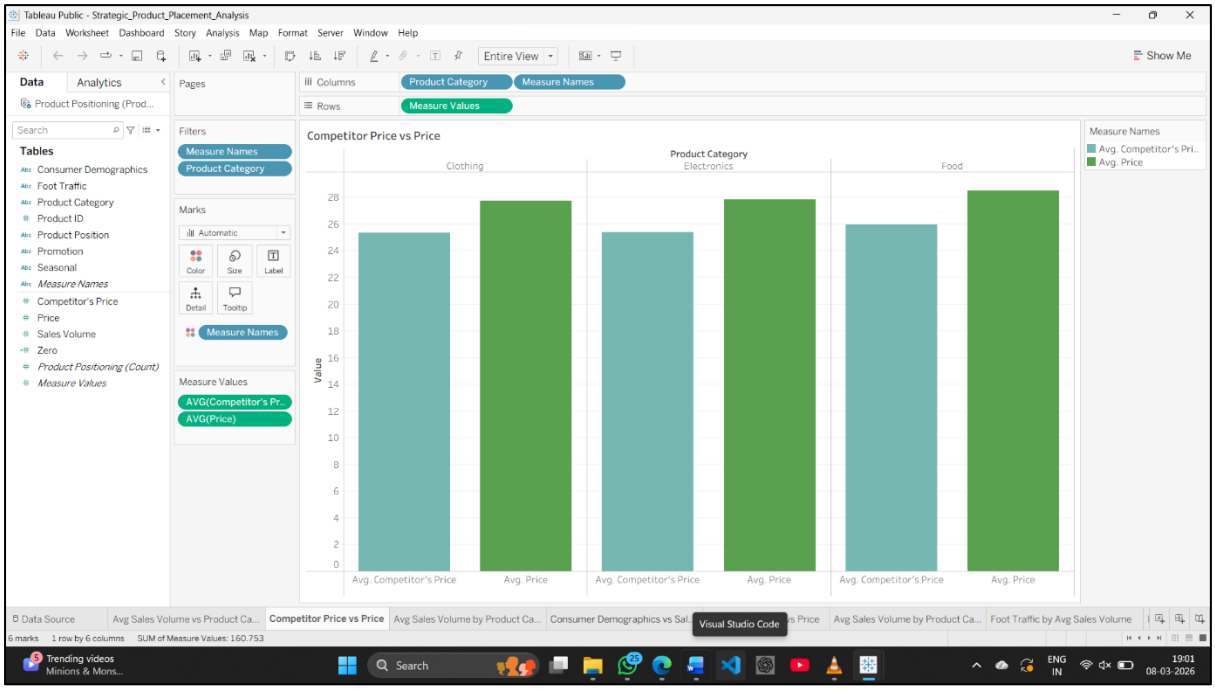


- **Data Preparation and Data Visualization**

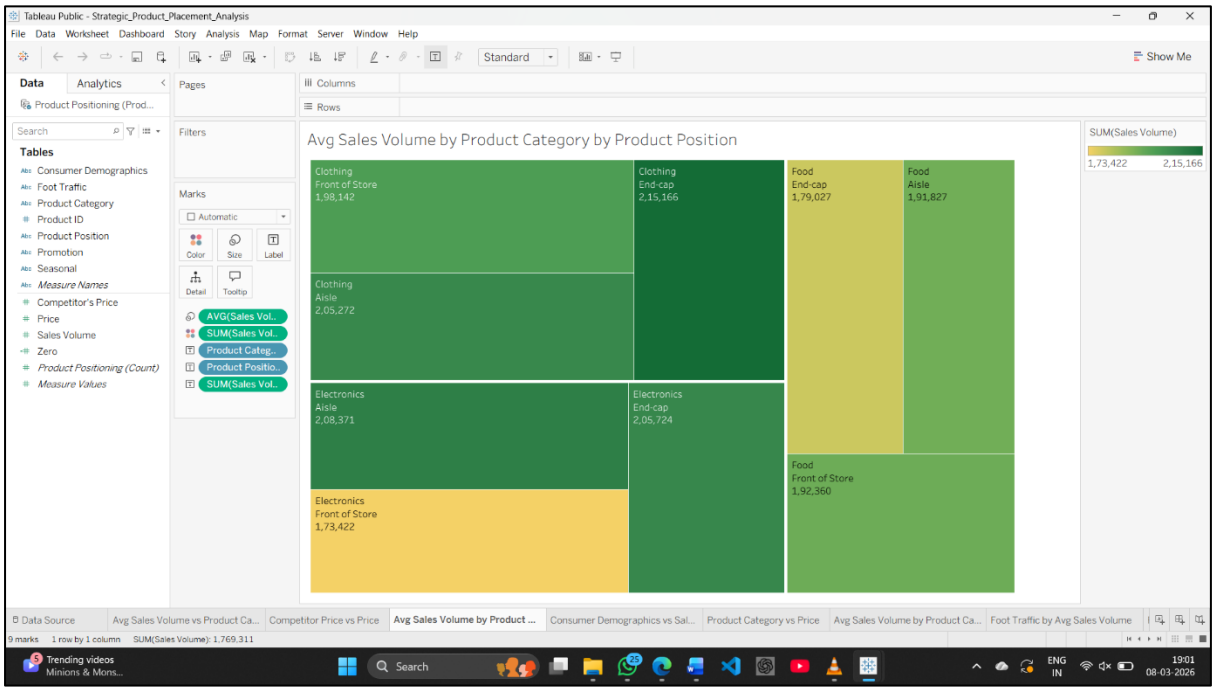
Sheet 1 - Avg Sales Volume vs Product Category



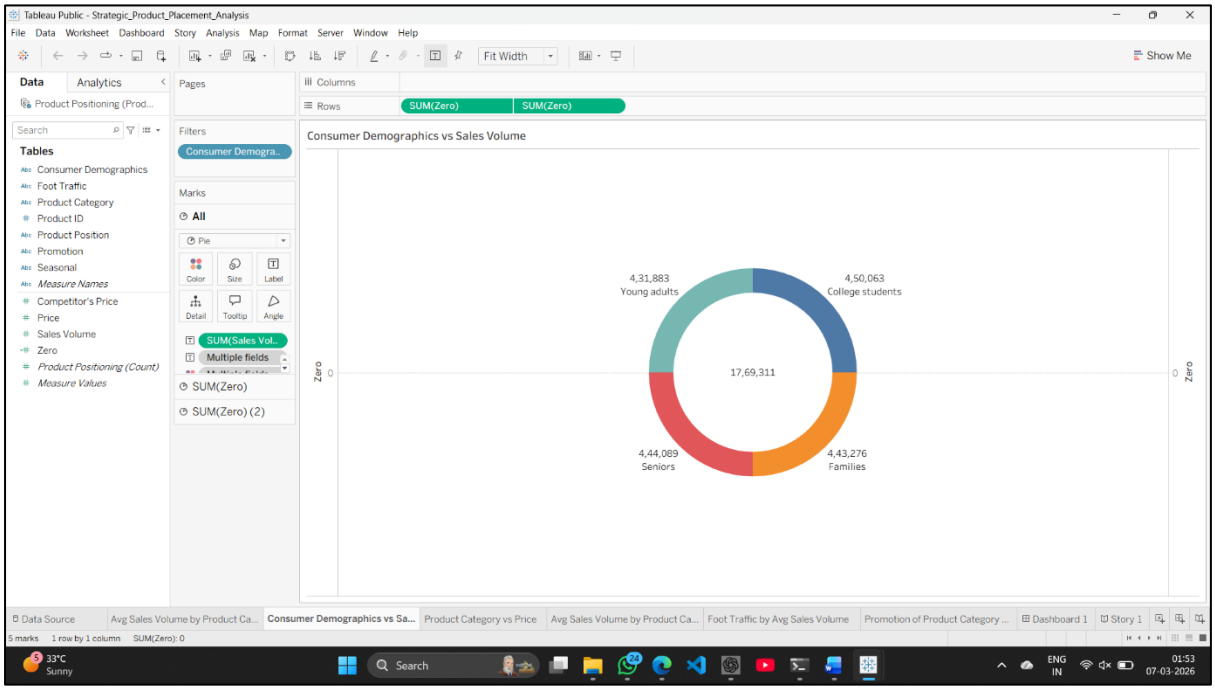
Sheet 2 - Competitor Price vs Price



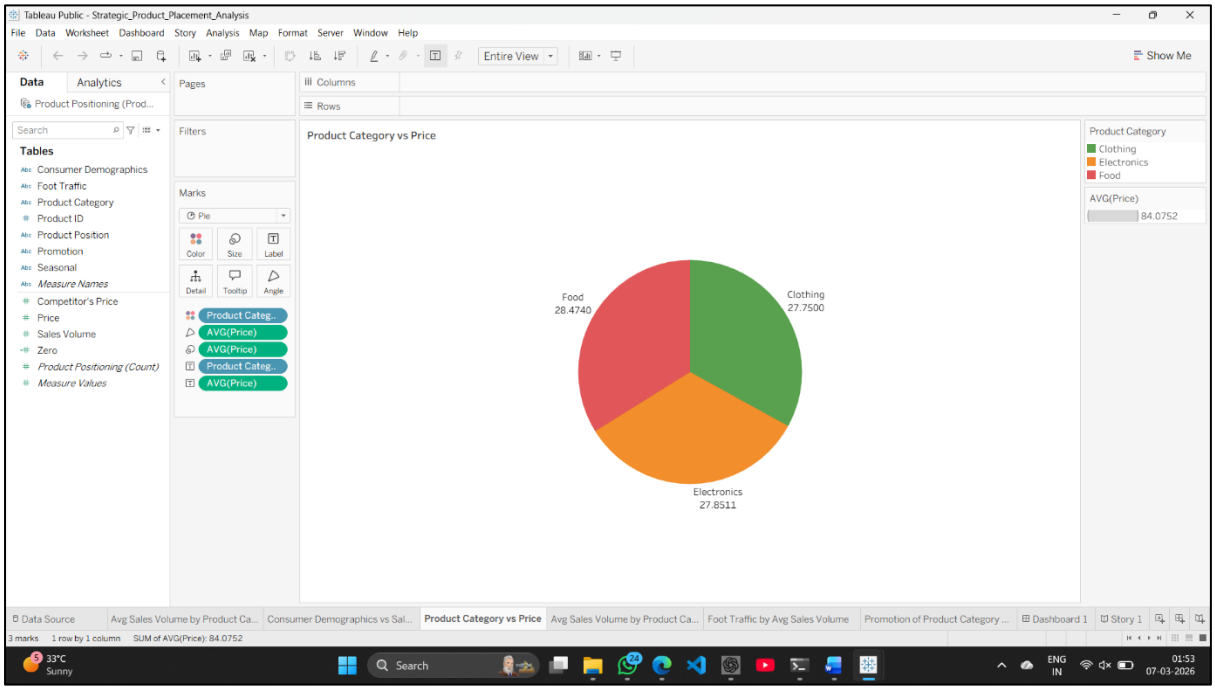
Sheet 3 - Avg Sales Volume by Product Category by Product Position



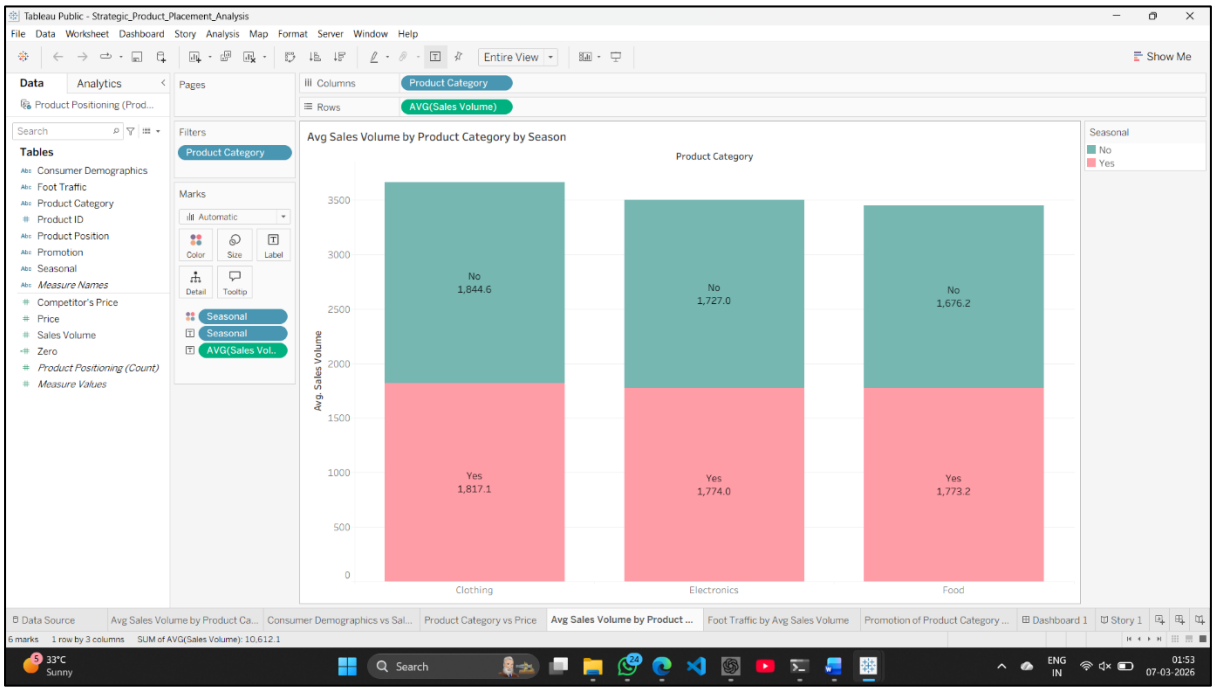
Sheet 4 - Consumer Demographics vs Sales Volume



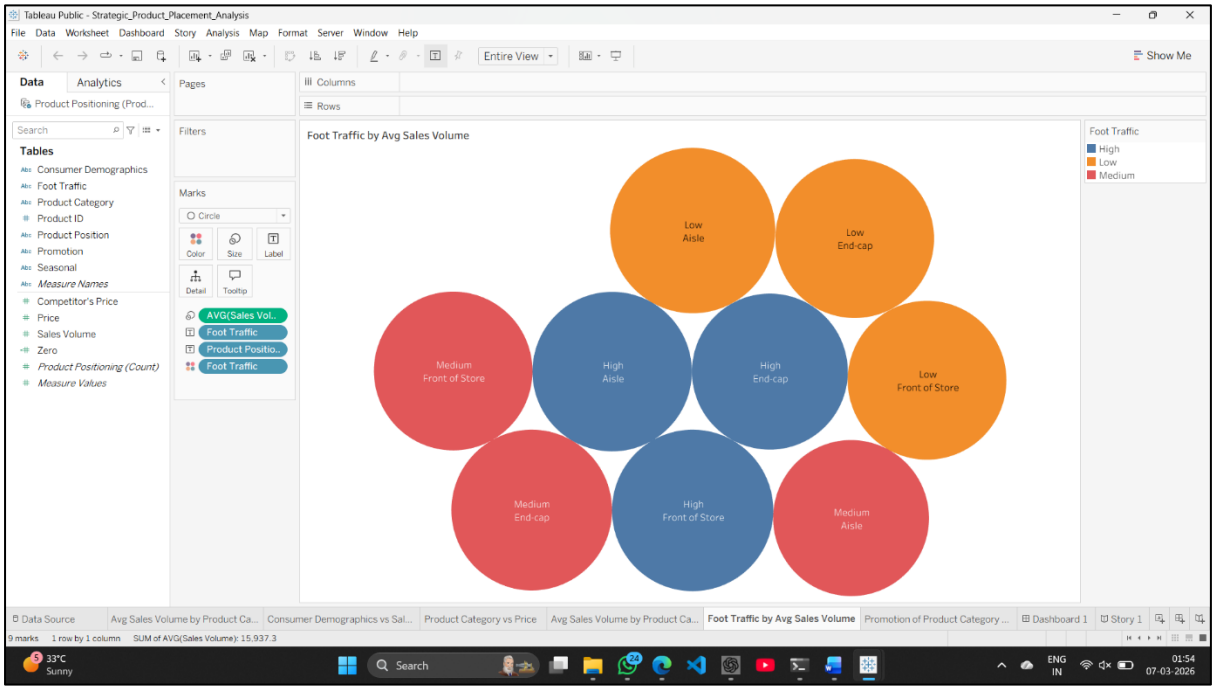
Sheet 5 - Product Category vs Price



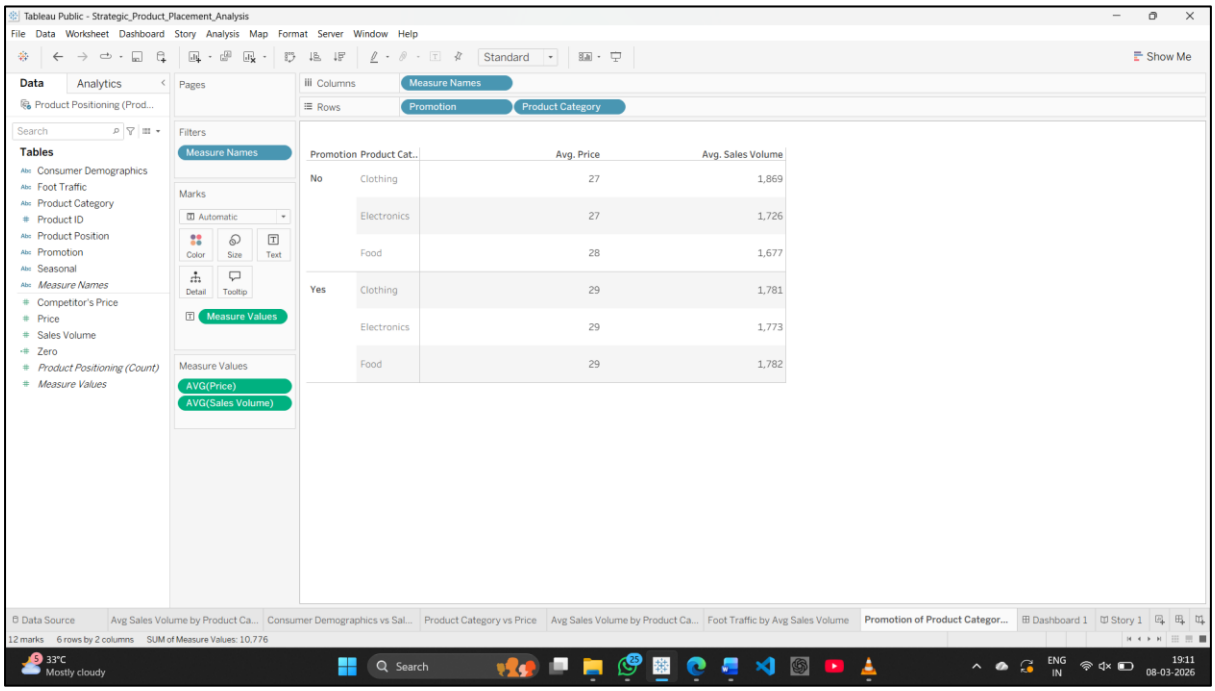
Sheet 6 - Avg Sales Volume by Product Category by Season



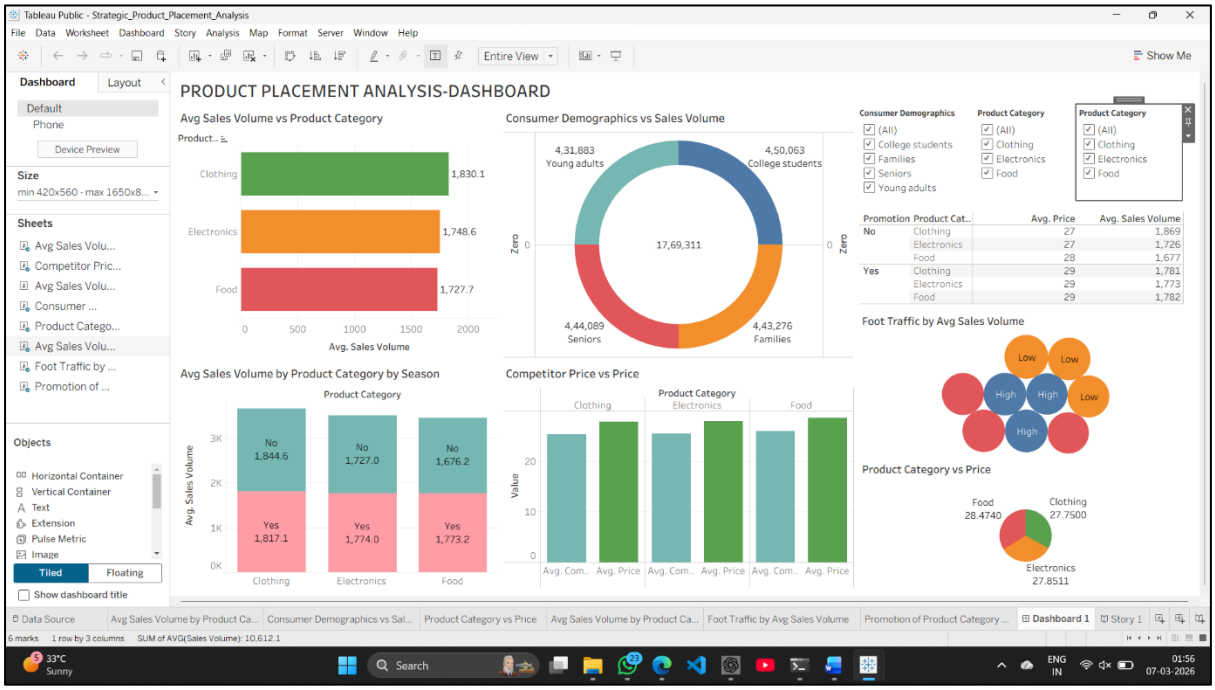
Sheet 7 - Foot Traffic by Avg Sales Volume



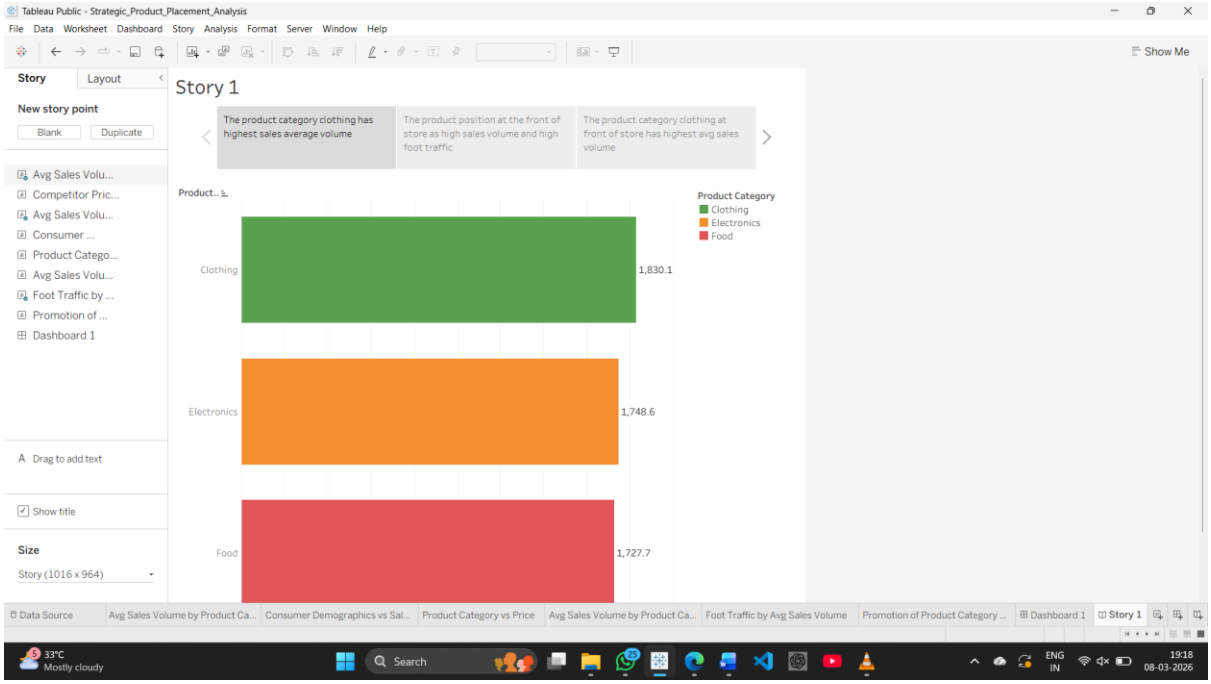
Sheet 8 - Promotion of Product Category on Price and Sales Volume



- **Design of Dashboard**



- **Scenes of Story**



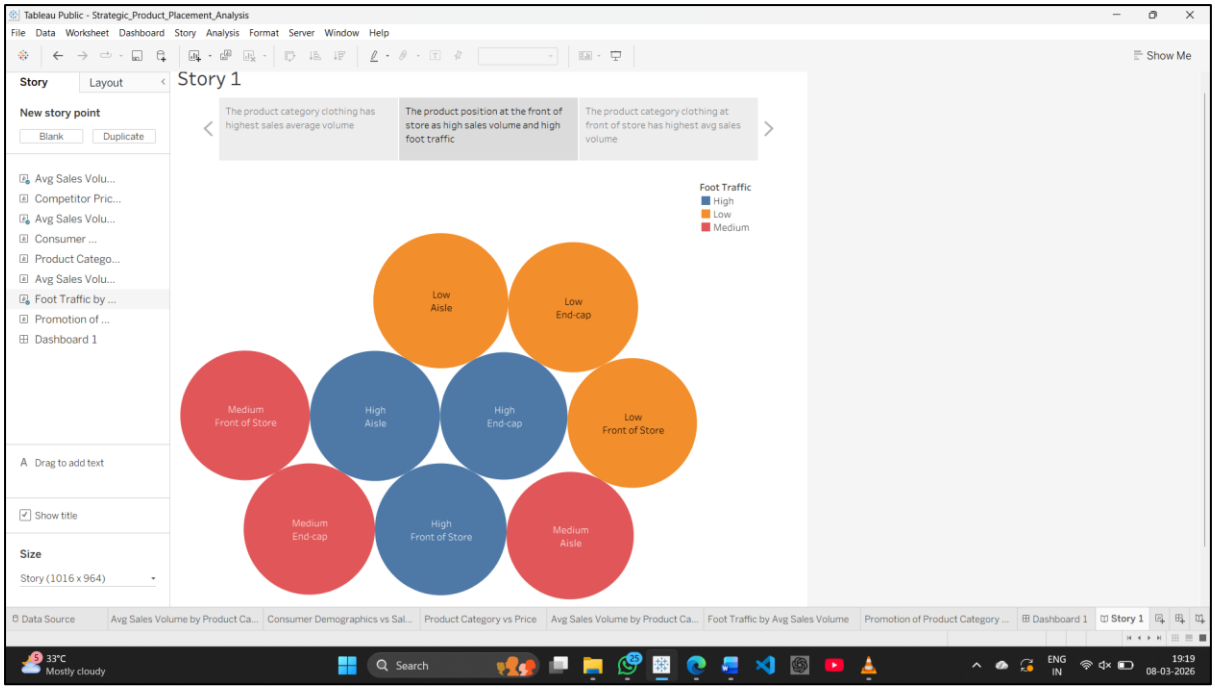
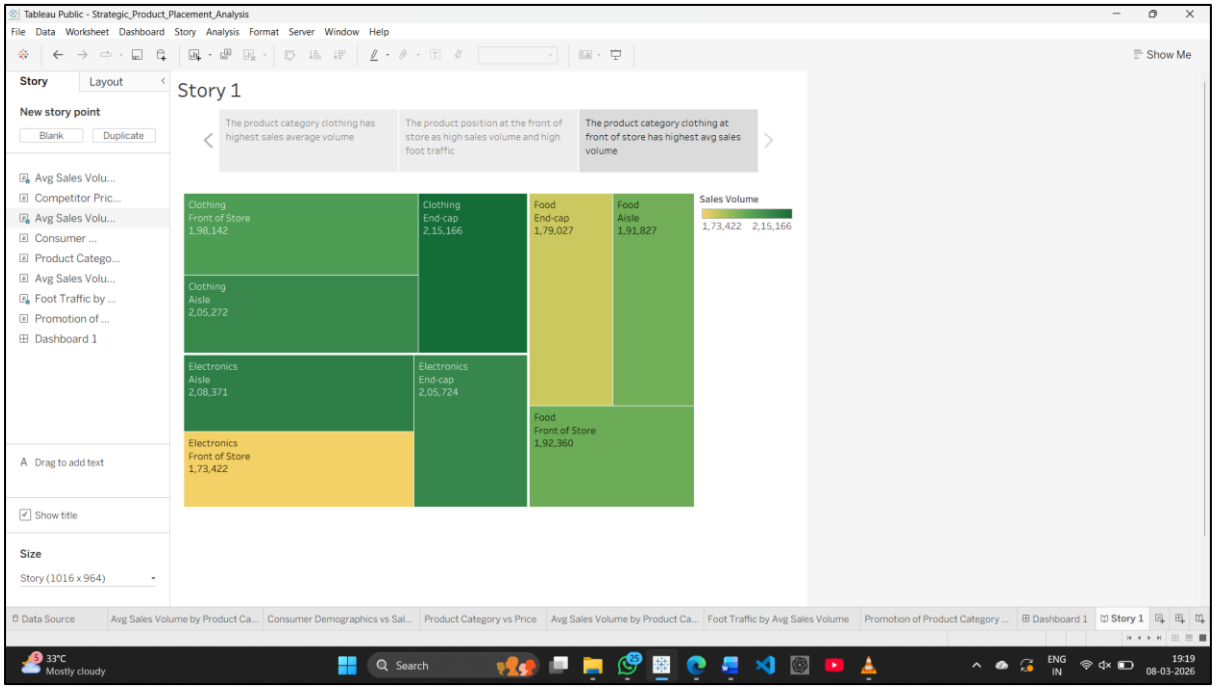


Tableau Public - Strategic_Product_Placement_Analysis
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Show Me
Story
Layout
New story point
Blank
Duplicate
Avg Sales Volu...
Competitor Pric...
Avg Sales Volu...
Consumer ...
Product Catego...
Avg Sales Volu...
Foot Traffic by ...
Promotion of ...
Dashboard 1
Drag to add text
Show title
Size
Story (1016 x 964)
Story 1
The product category clothing has highest sales average volume
The product position at the front of store as high sales volume and high foot traffic
The product category clothing at front of store has highest avg sales volume
Foot Traffic
High
Low
Medium
Low Aisle
Low End-cap
Medium Front of Store
High Aisle
High End-cap
Low Front of Store
Medium End-cap
High Front of Store
Medium Aisle
Data Source
Avg Sales Volume by Product Ca...
Consumer Demographics vs Sal...
Product Category vs Price
Avg Sales Volume by Product Ca...
Foot Traffic by Avg Sales Volume
Promotion of Product Category ...
Dashboard 1
Story 1
33°C
Mostly cloudy
Search
ENG
IN
19:19
08-03-2026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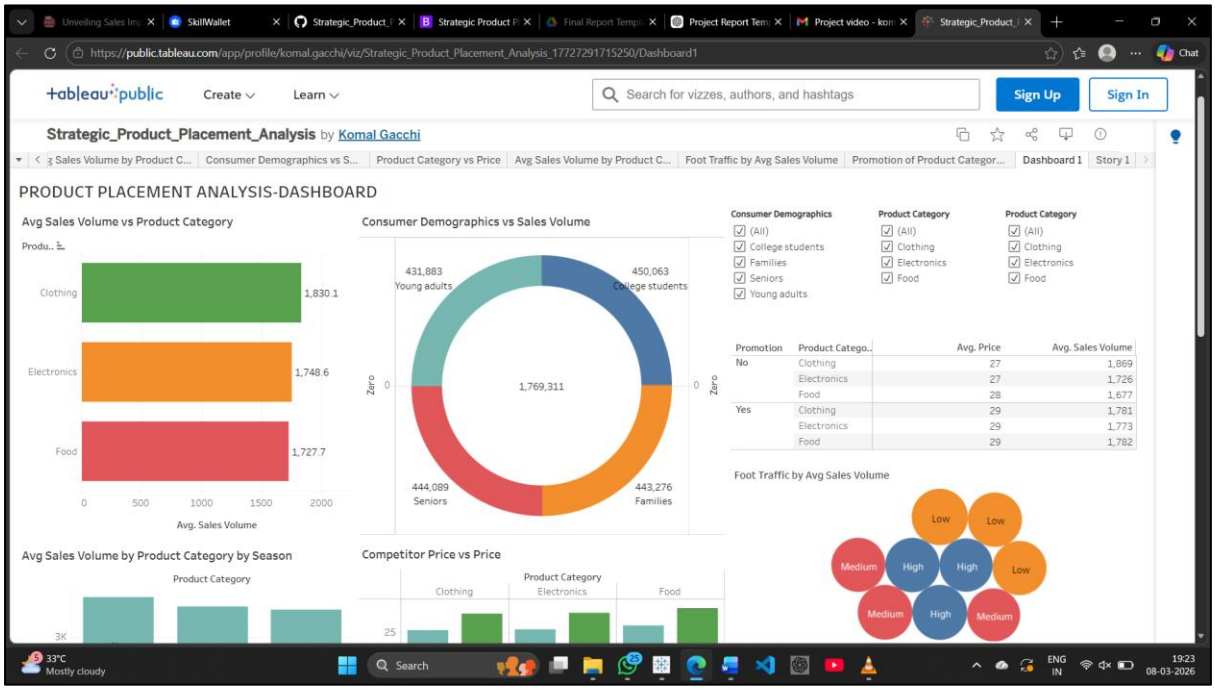
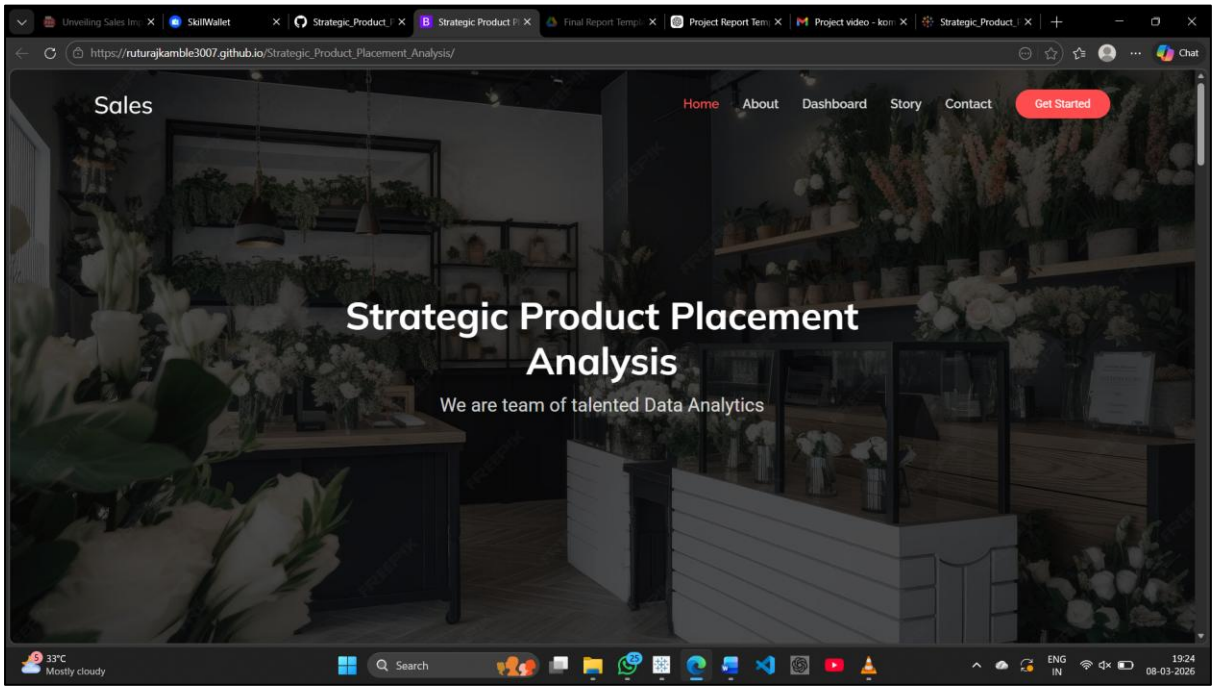
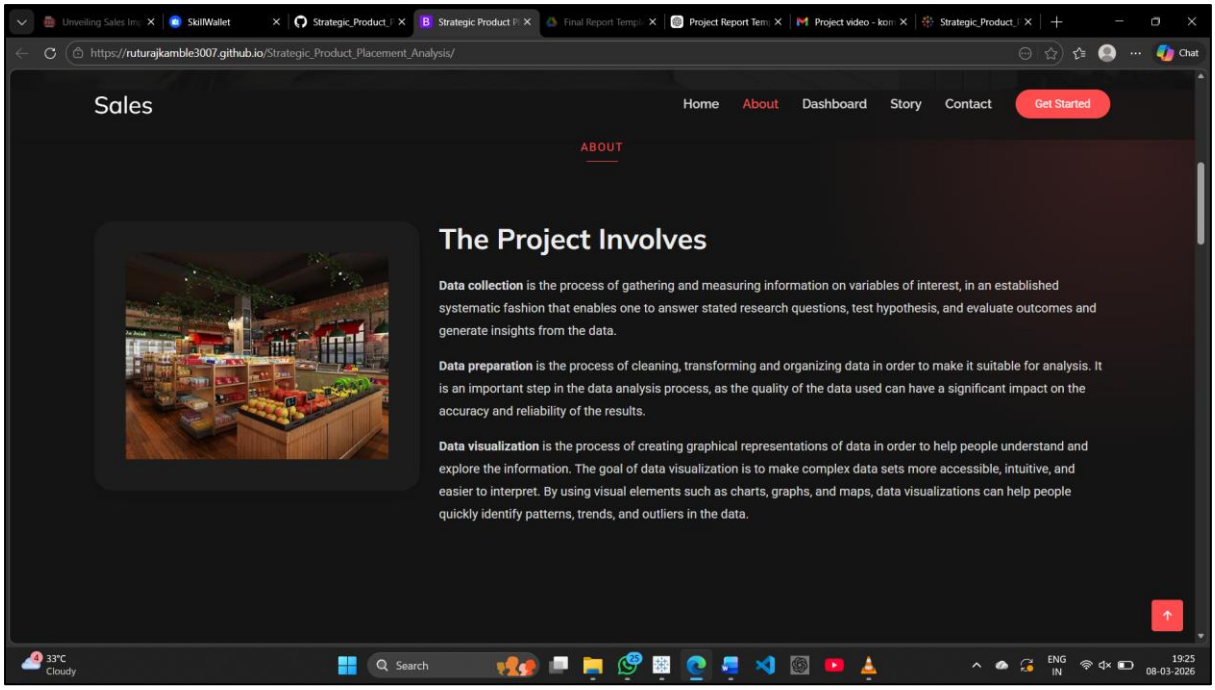
Sales Volume
1,73,422
2,15,166
Clothing Front of Store 1,98,142
Clothing End-cap 2,15,166
Food End-cap 1,79,027
Food Aisle 1,91,827
Clothing Aisle 2,05,272
Electronics Aisle 2,08,371
Electronics End-cap 2,05,724
Food Front of Store 1,92,360
Electronics Front of Store 1,73,422
Data Source
Avg Sales Volume by Product Ca...
Consumer Demographics vs Sal...
Product Category vs Price
Avg Sales Volume by Product Ca...
Foot Traffic by Avg Sales Volume
Promotion of Product Category ...
Dashboard 1
Story 1
33°C
Mostly cloudy
Search
ENG
IN
19:19
08-03-2026

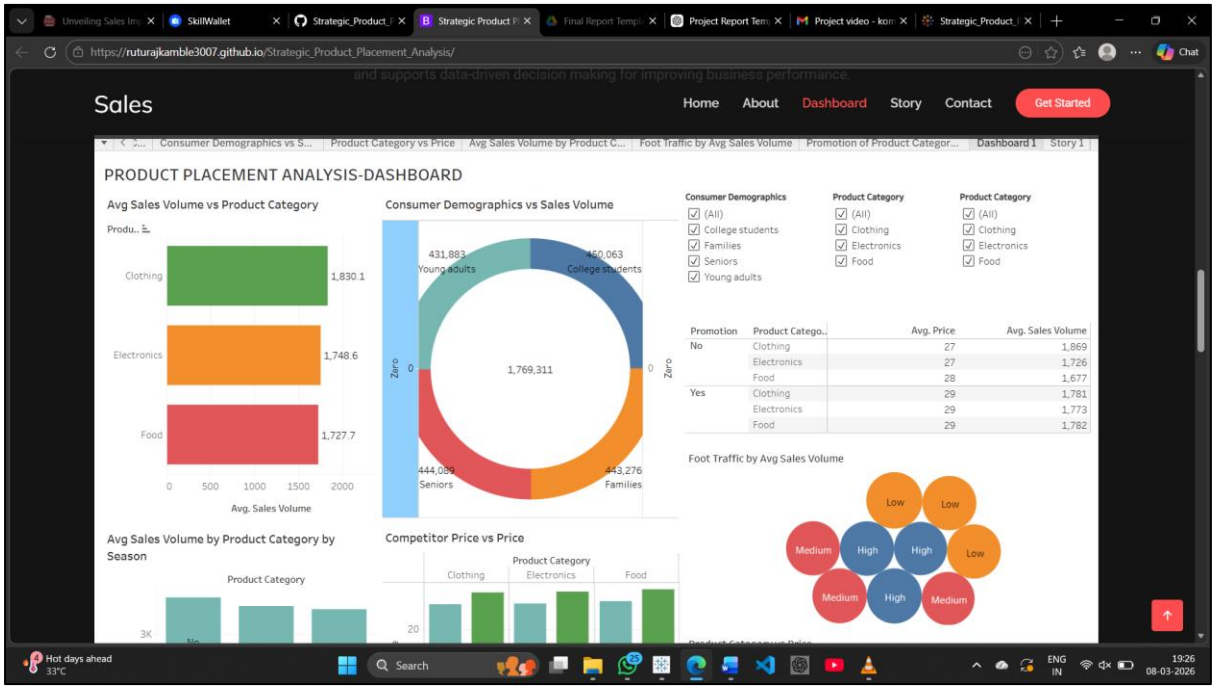
- **Publishing**



- **Dashboard and Story Embed with UI with Flask**




https://ruturajkamble3007.github.io/Strategic_Product_Placement_Analysis/
Sales
Home
About
Dashboard
Story
Contact
Get Started
ABOUT
The Project Involves
Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypothesis, and evaluate outcomes and generate insights from the data.
Data preparation is the process of cleaning, transforming and organizing data in order to make it suitable for analysis. It is an important step in the data analysis process, as the quality of the data used can have a significant impact on the accuracy and reliability of the results.
Data visualization is the process of creating graphical representations of data in order to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.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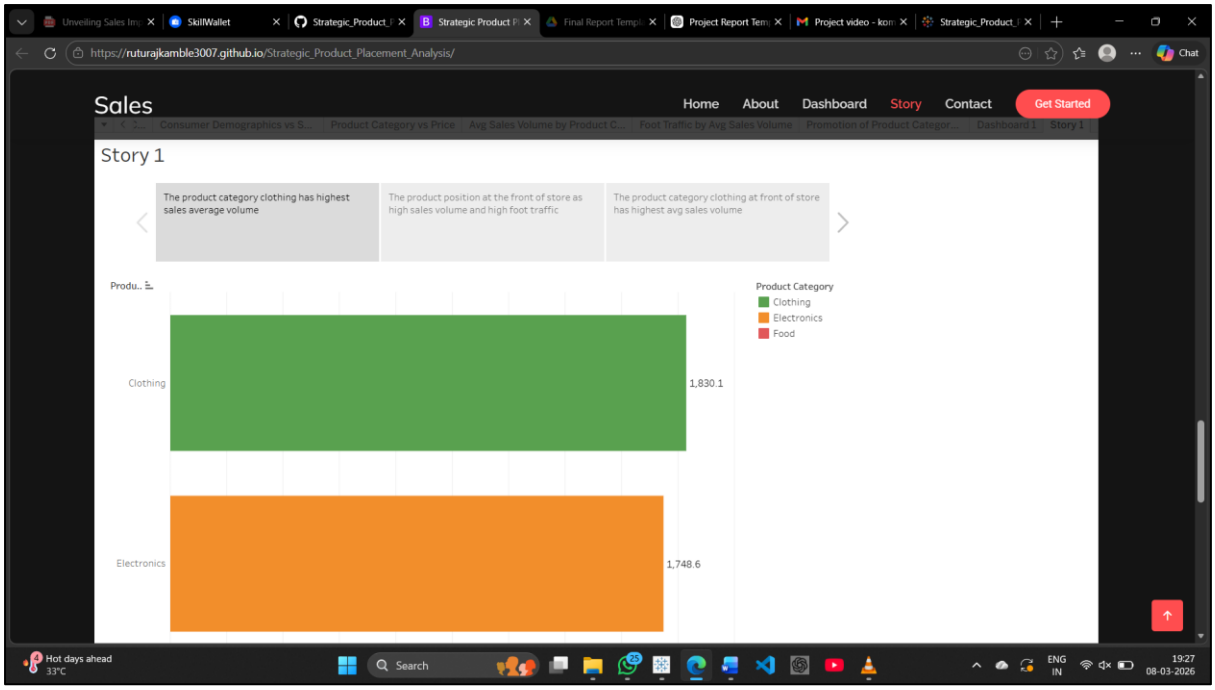
PRODUCT PLACEMENT ANALYSIS-DASHBOARD
Avg Sales Volume vs Product Category
Clothing
1,830.1
Electronics
1,748.6
Food
1,727.7
Avg. Sales Volume
Consumer Demographics vs Sales Volume
431,883
Young adults
450,063
College students
1,769,311
444,089
Seniors
443,276
Families
Consumer Demographics
(All)
College students
Families
Seniors
Young adults
Product Category
(All)
Clothing
Electronics
Food
Product Category
(All)
Clothing
Electronics
Food
Promotion
Product Catego..
Avg. Price
Avg. Sales Volume
No
Clothing
27
1,869
Electronics
27
1,726
Food
28
1,677
Yes
Clothing
29
1,781
Electronics
29
1,773
Food
29
1,782
Foot Traffic by Avg Sales Volume
Low
Low
Medium
High
High
Low
Medium
High
Medium
Avg Sales Volume by Product Category by Season
Product Category
Competitor Price vs Price
Product Category
Clothing
Electronics
Food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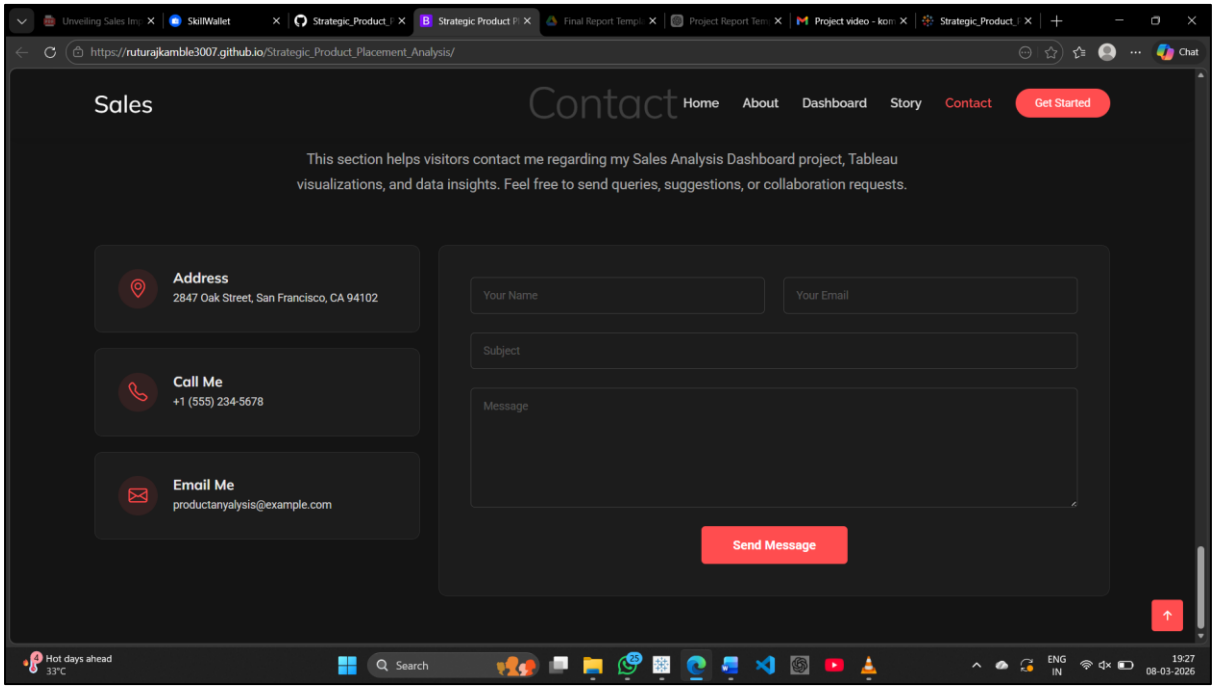
Electronics
Food
Clothing
1,830.1
Electronics
1,748.6


https://ruturajkamble3007.github.io/Strategic_Product_Placement_Analysis/
Sales
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This section helps visitors contact me regarding my Sales Analysis Dashboard project, Tableau visualizations, and data insights. Feel free to send queries, suggestions, or collaboration requests.
Address
2847 Oak Street, San Francisco, CA 94102
Call Me
+1 (555) 234-5678
Email Me
productanyalysis@example.com
Your Name
Your Email
Subject
Message
Send Message

ADVANTAGES & DISADVANTAGES

Advantages

- Provides clear visualization of complex sales data.
- Helps businesses identify profitable products.
- Supports better strategic decision-making.
- Interactive dashboards make analysis easier.
-

Disadvantages

- Insights depend heavily on the quality of data.
- Limited analysis if dataset is small or incomplete.
- Requires knowledge of Tableau for advanced analysis.

CONCLUSION

The Strategic Product Placement Analysis project demonstrates the importance of data visualization in understanding sales performance. By using Tableau dashboards, businesses can easily identify patterns in sales data and evaluate the effectiveness of product placement strategies.

The project shows how visual analytics can transform raw data into meaningful insights that support data-driven decision-making and improve overall business performance.

FUTURE SCOPE

Future improvements can enhance the capabilities of this project. The system can be integrated with real-time sales databases to provide up-to-date insights.

Machine learning techniques can also be applied to predict future sales trends and recommend optimal product placement strategies.

Additionally, the project can be extended by developing web-based dashboards or integrating it with business intelligence platforms for more advanced analytics.

APPENDIX

- **Kaggle Sales Dataset:**

https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales

- **Tableau Public Dashboard:**

https://public.tableau.com/app/profile/anuja.patil5853/viz/Product_Placement_Analysis_17731246739760/Story1

- **GitHub Repository:**

- **Demo Video:**